Topic: Autonomous equations and equilibrium solutions

Question: Find any equilibrium solutions of the autonomous differential equation, then determine whether each solution is stable, unstable, or semi-stable.

$$\frac{dy}{dt} = 4y - 4y^2$$

Answer choices:

A $y = 0$ is a stable solution and $y = 1$ is an unstable solution

B $y = 0$ is an unstable solution and $y = 1$ is a stable solution

C $y = 0$ and $y = 1$ are unstable solutions

D $y = 0$ and $y = 1$ are stable solutions

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Solution: B

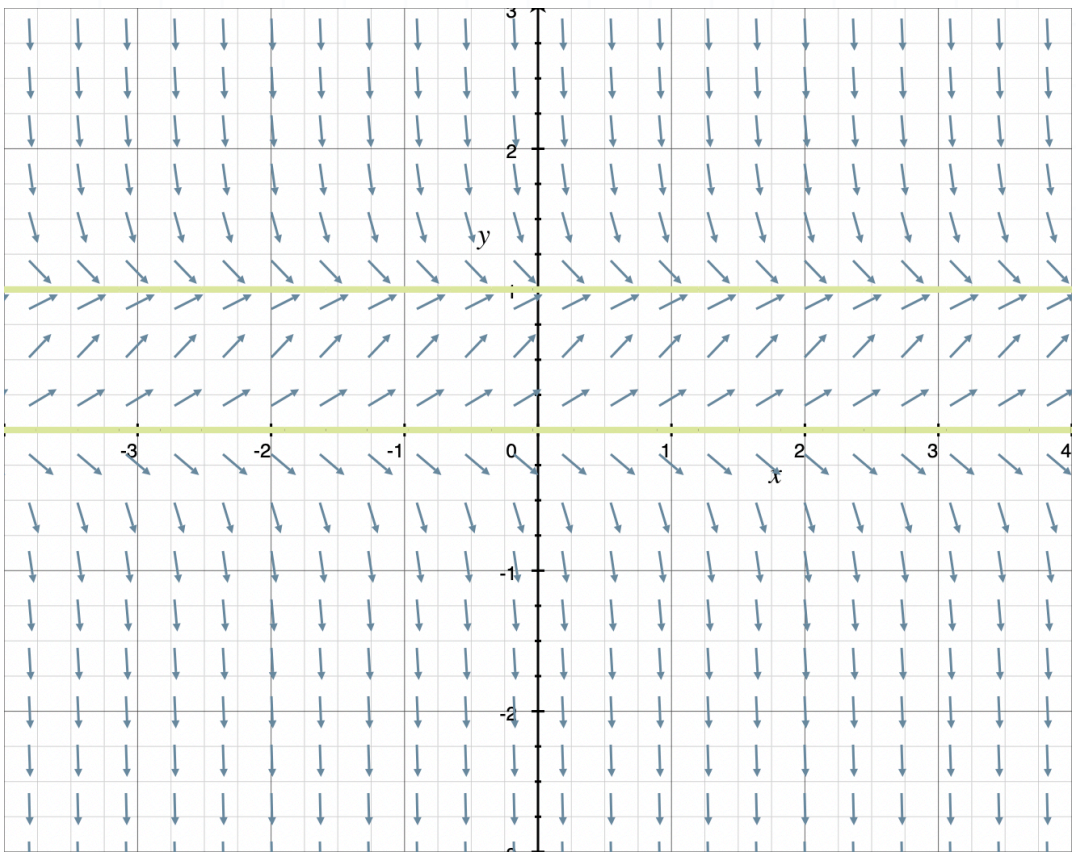
The autonomous differential equation has equilibrium solutions at

$$4y - 4y^2 = 0$$

$$y(1 - y) = 0$$

$$y = 0,\ 1$$

If we sketch the direction field and then overlay the equilibrium solutions, we get



These two equilibrium solutions divide the vertical axis into three intervals,

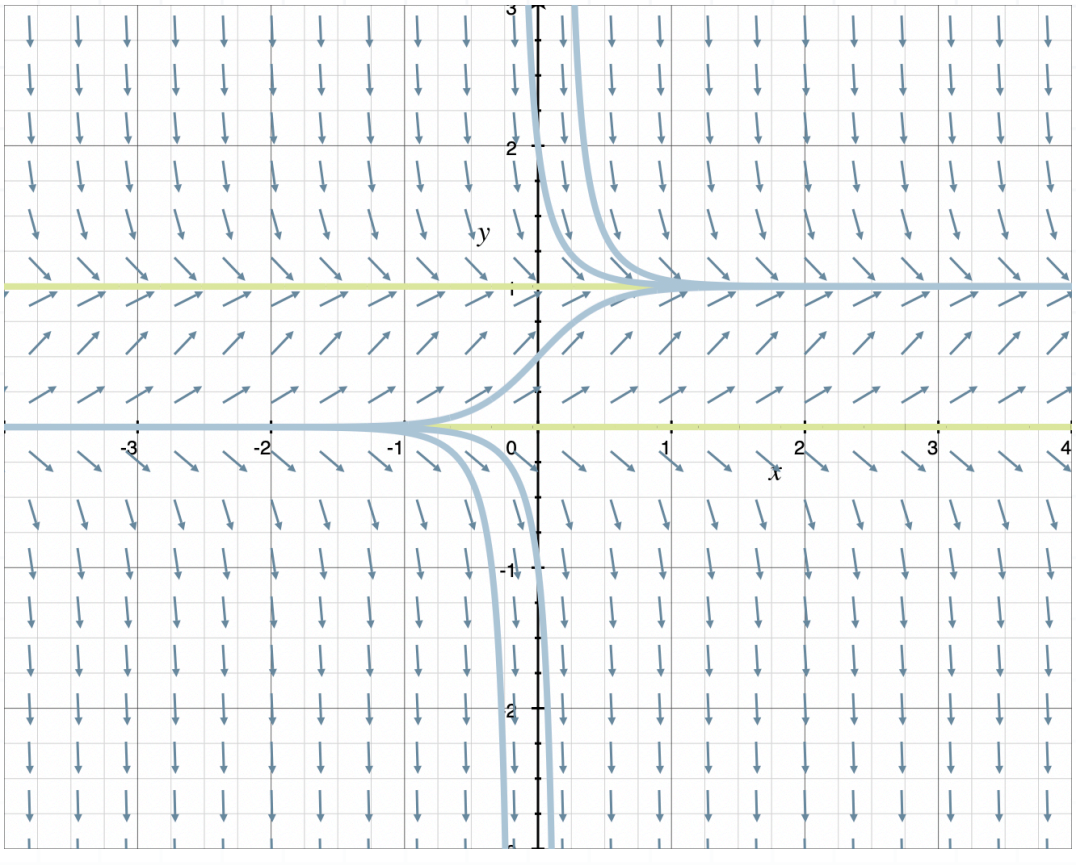
$$y < 0$$

$$0 < y < 1$$

$$1 < y$$

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We can use the direction field to sketch solution curves in each interval.



These solution curves approach $y = 1$ on both sides, but move away from $y = 0$ on both sides. Therefore, $y = 1$ is a stable equilibrium solution, while $y = 0$ is an unstable equilibrium solution.

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Topic: Autonomous equations and equilibrium solutions

Question: Find any equilibrium solutions of the autonomous differential equation, then determine whether each solution is stable, unstable, or semi-stable.

$$\frac{dy}{dt} = y(y-2)$$

Answer choices:

A $y = 0$ and $y = 2$ are unstable solutions

B $y = 0$ and $y = 2$ are stable solutions

C $y = 0$ is a stable solution and $y = 2$ is an unstable solution

D $y = 0$ is an unstable solution and $y = 2$ is a stable solution

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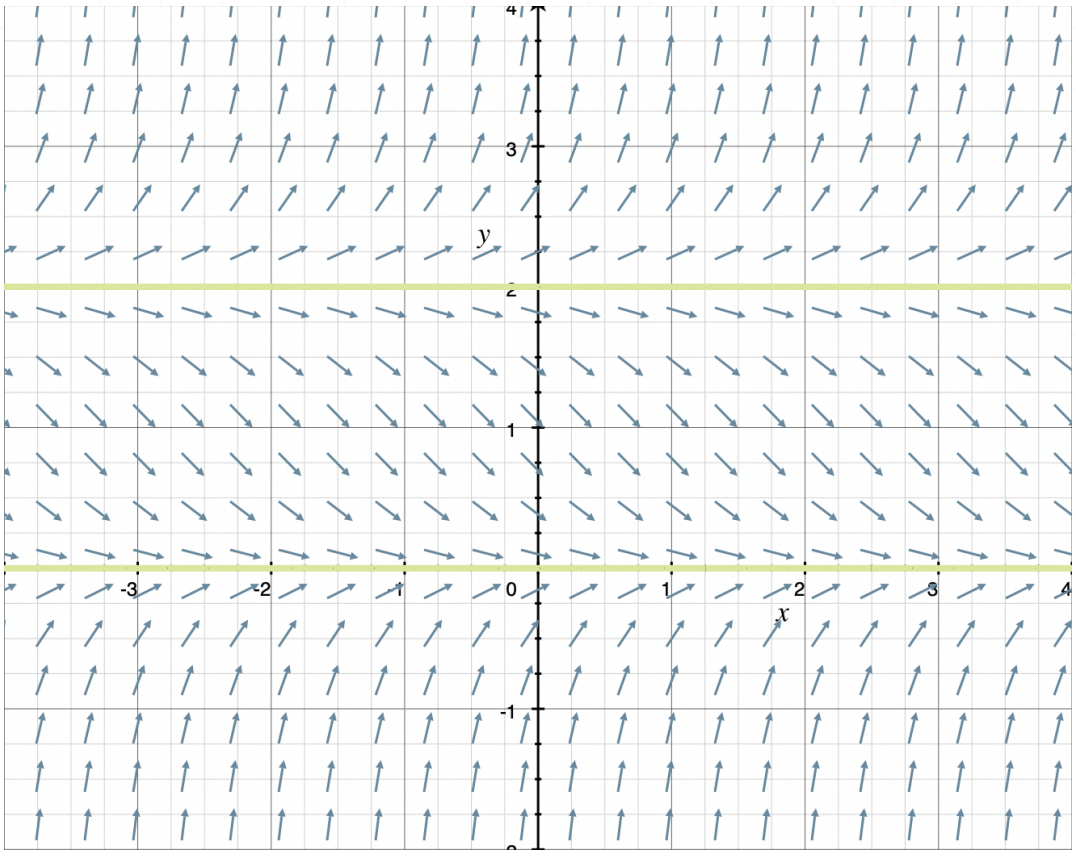
Solution: C

The autonomous differential equation has equilibrium solutions at

$$y(y-2)=0$$

$$y=0,\ 2$$

If we sketch the direction field and then overlay the equilibrium solutions, we get



These two equilibrium solutions divide the vertical axis into three intervals,

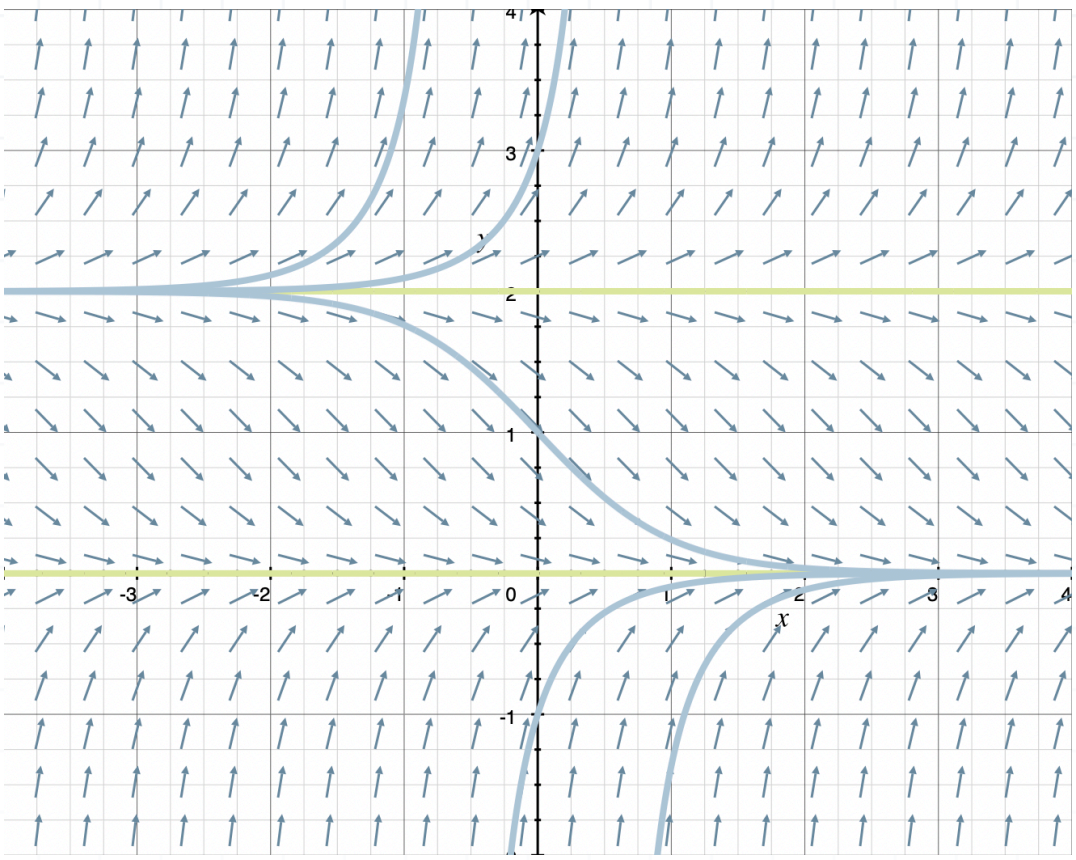
$$y<0$$

$$0<y<2$$

$$2<y$$

We can use the direction field to sketch solution curves in each interval.

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These solution curves approach $y = 0$ on both sides, but move away from $y = 2$ on both sides. Therefore, $y = 0$ is a stable equilibrium solution, while $y = 2$ is an unstable equilibrium solution.

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Topic: Autonomous equations and equilibrium solutions

Question: Find any equilibrium solutions of the autonomous differential equation, then determine whether each solution is stable, unstable, or semi-stable.

$$y' = y^3 - 5y^2 + 6y$$

Answer choices:

A $y = 0$, $y = 2$, and $y = 3$ are unstable solutions

B $y = 0$, $y = 2$, and $y = 3$ are stable solutions

C $y = 0$ and $y = 3$ are stable solutions and $y = 2$ is an unstable solution

D $y = 0$ and $y = 3$ are unstable solutions and $y = 2$ is stable solution

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Solution: D

The autonomous differential equation has equilibrium solutions at

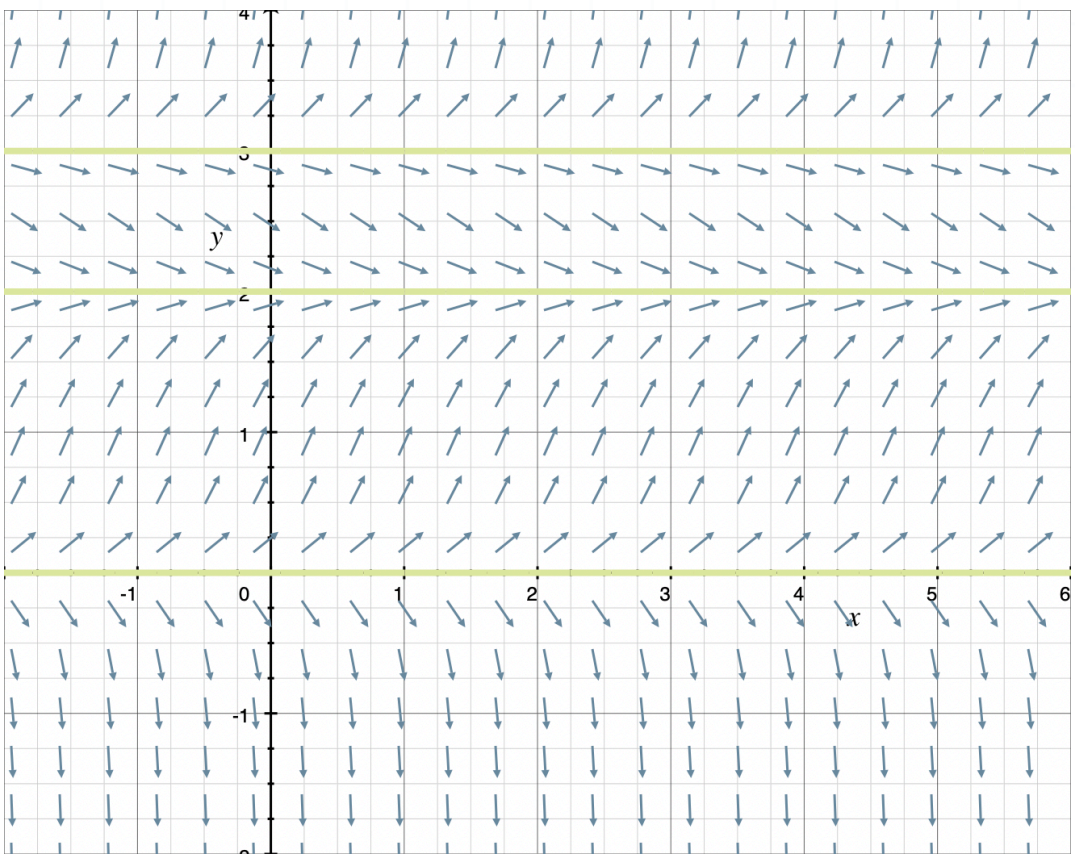
$$y^3 - 5y^2 + 6y = 0$$

$$y(y^2 - 5y + 6) = 0$$

$$y(y - 2)(y - 3) = 0$$

$$y = 0,\ 2,\ 3$$

If we sketch the direction field and then overlay the equilibrium solutions, we get



These three equilibrium solutions divide the vertical axis into four intervals,

$$y < 0$$

$$0 < y < 2$$

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$$2 < y < 3$$

$$3 < y$$

and we can see from the direction field that solution curves will approach $y = 2$ on both sides, but move away from $y = 0$ and $y = 3$ on both sides. Therefore, $y = 2$ is a stable equilibrium solution, while $y = 0$ and $y = 3$ are unstable equilibrium solutions.

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